

Assignment 4- Solutions

⇒ Design loads

$$X = 412$$

$$T^* = \frac{X}{3} = 137 \text{ kNm}$$

$$V^* = \frac{400}{3} = 549 \text{ kN}$$

$$M^* = \frac{800}{3} \text{ kNm}$$

⇒ Dimensions related to external closed fitment

assume N12 stirrups, cover = 30mm

$$x = 900 - 2 \times (30 + 12/2) = 828 \text{ mm}$$

$$y = 700 - 2 \times (30 + 12/2) = 628 \text{ mm}$$

$$U_h = 2 \times (628 + 828) = 2912 \text{ mm}$$

$$A_h = 828 \times 628 = 520 \times 10^3 \text{ mm}^2 : \text{Area enclosed by centreline of closed fitment.}$$

⇒ Dimensions related to the shear flow path

$$x = 900 - \frac{120}{2} - \frac{120}{2} = 780 \text{ mm}$$

$$y = 700 - \frac{120}{2} - \frac{150}{2} = 565 \text{ mm}$$

$$U_o = 2 \times (780 + 565) = 2690 \text{ mm}$$

$$A_o = 780 \times 565 = 440700 \text{ mm}^2 : \text{Area enclosed by shear flow path.}$$



⇒ check if torsional effects are considered.

$$T_{cr} = 0.33 \sqrt{f'_c} \frac{A_{cp}^2}{U_c} = 0.33 \sqrt{40} \times \frac{436000^2}{4400} = 90.25 \text{ kNm}$$

$$A_{cp} = (900 \times 550) - (660 \times 430) + (1500 \times 150) = 436200 \text{ mm}^2$$

$$U_c = 900 + (550 \times 2) + 1500 + (150 \times 2) + 600 = 4400 \text{ mm}$$

$$0.25 T_{cr} = 0.25 \times 0.75 \times 90.25 = 16.9 \text{ kNm} < T^*$$

torsional effects shall be considered.

(0.5 points)

* Design for flexure (1 point)

assume $Z_{est} = 0.95d = 0.95 \times 642 = 609.9 \text{ mm}$

$d = 700 - 30 - 12 - 3^2/2 = 642 \text{ mm}$

- assuming under-reinforced ^(u/r) $\phi = 0.85$

$$T = A_{st} f_{sy} = \frac{M^*}{\phi Z_{est}} = \frac{800 \times 10^6}{0.85 \times 609.9} = 1543.17 \text{ kN}$$

$$A_{st, \text{required}} = \frac{1543.17 \times 10^3}{500} = 3086.33 \text{ mm}^2$$

$\therefore$ use 4N32 : $A_{st} = 3216 \text{ mm}^2$

assume neutral axis in flange:

$$\alpha_2 = 0.85 - 0.0015 \times 40 = 0.79$$

$$\beta = 0.97 - 0.0025 \times 40 = 0.87$$

$$\delta K_u d = \frac{3216 \times 500}{0.79 \times 40 \times 1500} = 33.92 \text{ mm} \quad \text{N/A in flange}$$

$$K_u = \frac{33.92}{0.87 \times 642} = 0.061 < 0.545 \text{ u/r}$$

$< 0.36, \phi = 0.85 [16.2.2.2]$

$$M = \phi A_{st} f_{sy} \left(d - \frac{1}{2} \delta K_u d \right)$$

$$= 0.85 \times 3216 \times 500 \times (642 - 0.5 \times 33.92) \left(\times 10^{-6} \right)$$

$$= 843.3 \text{ kNm} > M^* \quad \therefore \text{OK}$$

A Design for Shear (1.5 points)

using the simplified method [C18.2 4.3]

* Find V_{uc} [C18.2.4.1]

$$V_{uc} = K_u b_v d_v \sqrt{f_c}$$

$$= 0.15 \times 240 \times 578 \times \sqrt{40}$$

$$= 131.601 \text{ kN}$$

$$b_v = 120 \times 2 = 240 \text{ mm}$$

$$d_v = \max 0.72D \text{ or } 0.9d.$$

$$0.72 \times 700 \text{ or } 0.9 \times 642 = 578 \text{ mm}$$

$$\phi V_{uc} = 0.75 \times 131.6 = 98.7 \text{ kN}$$

$$\phi(V_{uc} + V_{us}) \geq V^* \rightarrow V_{us} \geq \frac{V^*}{\phi} - V_{uc}$$

$$\cot 36^\circ = 1.376$$

$$\phi V_{us} = V^* - \phi V_{uc} = 549 - 98.7 = 450.3 \text{ kN}$$

$$\frac{A_{sv}}{s} \geq \frac{V_{us}}{f_{sy} f d_v \cot \theta_v} = \frac{450.3 \times 10^3}{500 \times 578 \times \cot 36^\circ} = 1.51 \text{ mm}^2/\text{mm}$$

- check for $A_{sv, \min}$ [C18.2.1.7]

$$\frac{A_{sv, \min}}{s} = \frac{0.08 \sqrt{f_c} b_v}{f_{sy} f} = \frac{0.08 \times \sqrt{40} \times 240}{500} = 0.24 \text{ mm}^2/\text{mm}$$

$$< (1.51)$$

- check for web crushing [C18.2.3.3]

$$V_{u, \max} = 0.55 \times 0.9 \times 40 \times 240 \times 578 \times \left(\frac{\cot 36^\circ + \cot 90^\circ}{1 + \cot^2 36^\circ} \right)$$

$$= 1306.11 \text{ kN}$$

- check $t_w \leq A_{ch} / u_n = \frac{570 \times 10^3}{2912} = 178 \text{ mm} > t_w = 120 \text{ mm}$ ok

$$\frac{V^*}{b_v d_v} + \frac{T^*}{1.7 t_w A_{ch}} \leq \frac{\phi V_{u, \max}}{b_v d_v} \quad \text{cl 8.2.3.4}$$

$$\frac{549 \times 10^3}{240 \times 578} + \frac{137 \times 10^6}{1.7 \times 120 \times 570 \times 10^3} \leq \frac{0.7 \times 1306 \times 10^3}{240 \times 578}$$

$$3.46 + 1.29 \leq 6.59$$

$5.25 \leq 6.59 \therefore \text{OK for web crushing}$

- additional force in longitudinal bars
cl 8.2.7

$$\Delta F_{eds} = 0.5 (V^* + \phi V_u) \cot \theta_v$$

$$= 0.5 (549 \times 10^3 + 98.7 \times 10^3) \cot 36^\circ \times 10^{-3}$$

$$\text{or } \min = 445.6 \text{ kN}$$

$$= V^* \cot \theta_v = 755 \text{ kN}$$

$$A_{sv} = \frac{\Delta F_{eds}}{0.85 \times f_{yv}} = \frac{445.6 \times 10^3}{0.85 \times 500} = 1048 \text{ mm}^2$$

* Design for Torsion (1 point)

$$\left[\begin{array}{l} \text{cl 8.2.3.1} : \phi T_{us} \geq T^* \\ \text{cl 8.2.5.6} : T_{us} = 2 A_o \frac{A_{sw} f_{yv} \cot \theta_v}{s} \end{array} \right.$$

$$\Rightarrow \frac{A_{sw}}{s} = \frac{T^*}{\phi 2 A_o f_{yv} \cot \theta_v}$$

$$= \frac{137 \times 10^6}{0.75 \cdot 2 \times 440700 \times 500 \times 1.376}$$

$$= 0.301 \text{ mm}^2/\text{mm}$$

- additional force in longitudinal bars [cl 8.2.7]

$$\Delta F_{edt} = 0.5 T^* \frac{u_o}{2 A_o} \cot \theta_v \quad (7/0)$$

$$= 0.5 \times 137 \times 10^6 \times \frac{2690}{2 \times 440700} \times 1.376 = 288 \text{ kN}$$

$$A_{s,t} = \frac{F_t d_t}{0.85 \times f_{sy}} = \frac{288 \times 10^3}{0.85 \times 500} = 677.6 \text{ mm}^2$$

* Final design (1.5 points)

→ bottom longitudinal reo. to carry the combined force due to M^*, V^*, T^*

$$A_{st} + A_{s,v} + A_{s,t} = 3068 + 1048 + 678 = 4794 \text{ mm}^2$$

$$\therefore \text{use } 7 \text{ N32} = 7 \times 804 = 5628 \text{ mm}^2$$

$$\text{centre-to-centre spacing} = \frac{900 - 30 \times 2 - 12 \times 2 - 32}{6}$$

↳ 1 bar of N32 (804 mm²) @ each bottom corner and evenly spaced throughout the bottom flange @ 130 mm spacing.

→ top longitudinal reo. to carry the combined force due to $M^*, V^*, T^* < 0$

$$A_{s,v} + A_{s,t} - \underbrace{A_{st}} = 1048 + 678 - 3068 = -1342 < 0$$

∴ provide minimum torsional reinforcement in the top flange

→ transverse reo. due to $V^* \& T^*$:

choose 2-leg M12 closed stirrups → $A_{sv} = 2 \times 110 = 220 \text{ mm}^2$
 ↳ shear

required spacing:

$$\frac{1}{s} \geq \frac{A_{sv}/s}{A_{sv}} + \frac{A_{sw}/s}{A_{sw}} = \frac{4.51}{220} + \frac{0.301}{110} = 0.0096$$

$$s \leq 104 \approx 100 \text{ mm}$$

$$\begin{aligned} \text{max spacing} &= \min(0.12 U_h, 300 \text{ mm}) \\ &= 300 \text{ mm [Cl 8.3.3 b]} \\ &= \text{Choose 2-leg N12 closed stirrups @ 100 mm spacing} \end{aligned}$$

Part B

(i) cracking torque $\rightarrow T_{cr} = 0.33 \sqrt{f'_c} \frac{A_{cp}^2}{U_o} \sqrt{1 + \frac{\sigma_{cp}}{0.33 f'_c}}$ (0.5 points)

increase σ_{cp} (pre-compression) results in larger T_{cr} $\therefore$ more resistance to torsion cracking

(ii) angle of inclination (0.5 points) between initial cracks & the longitudinal axis

$\rightarrow$ when compression N^* is considered

ϵ_{xc} decreases and makes θ_v smaller according to Cl 8.2.4.

Cl 8.2.5.2 and Cl 8.2.5.6 for the same

$V_{us} + T_{us}$, smaller θ_v results in decrease of A_{sw}/s and A_{sv}/s

(iii) longitudinal & transverse req (0.5 points)
From Cl 8.2.7 increase in pre-compression results in increase in additional longitudinal force ΔF_{ed} , resulting in more longitudinal reinforcement required